FINAL PROJECTS SPECIFICATIONS

USE THE FOLLOWING IMPORTANT INFORMATION TO:

(i) Connect with your team
(ii) Reflect on your (Individual and/or collective) Final Project options and alternatives, based on the restrictions listed in the document uploaded here
(iii) Make a final decision regarding your selected topics
(iv) Put together a realistic work-plan
(v) Start working in your (Individual and collective) projects

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THE FINAL PROJECT, WORTH 60% OF THE FINAL GRADE OF THE COURSE, INCLUDES 2 PARTS:

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- THE INDIVIDUAL CONSOLIDATED ASSIGNMENT
- THE FINAL TEAM PROJECT

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If you are not in a team, as an individual you can either:

(i) Submit a Development Project yourself, or
(ii) Submit the Critical Review of a pair (2) of Papers from the list below.

…but PLEASE NOTICE that a team of 2 or more can only submit a Development Project (i.e., The Critical Review of a pair of papers is reserved to individuals and will not be allowed as a team project).

(READ THE SECTION TITLED "SUMMARY" BELOW FOR CLARIFICATION)

Regarding Research Papers, you'll find an extensive list of them in an attachment to the present document.

Regarding the Development Projects, you'll also find a list of possible topics below, but you can also propose your own idea to the instructors. Intellectually ambitious ideas that go beyond the original scope of this course, will be encouraged and given special consideration by the instructors in terms of resources and flexibility.

Regarding the Critical Review of a pair of research papers as a substitute to the Final Development Project (only applicable to an individual who does not belong to any group and feels unable to do any Development Project), the deliverables are: {(i) One self-contained PPP (>25) slides presentation for each paper and (ii) A monography (length at student's own discretion) summarizing each PP Presentation}. Note: please keep in mind that the Critical Review of a paper is not just a condensed digest, but a critical analysis of the paper where you discuss its pros. and cons., its relevance within the contemporary literature, its possible future developments in terms of new lines of research, literature that enriches or refutes it, suggest a complementary bibliography, etc.

Regarding the Final Development Project (Team Work), once your work is ready, you'll upload to QUERCUS (in the appropriate space reserved for that purpose called "FINAL DEVELOPMENT PROJECTS"), A FOLDER containing the following: {(i) Your Team's Power Point Presentation (>25 Slides), (ii) Commented code developed (if any) and (iii) A 1 or 2 pages Executive Summary of the work presented}. THE TEAM MEMBERS OFFICIAL NAMES AND U OF T ID #'s SHOULD BE CLEARLY VISIBLE IN ALL THE DELIVERABLES. (Obviously, if you are an individual with no team, you'll just upload {(i), (ii), (iii)} to your name).

Regarding the Individual Consolidating Assignment (Individual Work); this is for everybody: each student in the class will select 2 papers from the list below and will do a Critical Review of each paper. Also in this case, the deliverables for the Individual Consolidating Assignment are: {(i) A self-contained PP (>25 slides) Presentation for each paper and (ii) A monography (length at your own discretion) summarizing each Presentation}. Once your work is ready, you'll upload to QUERCUS (in the appropriate space reserved for that purpose called "INDIVIDUAL CONSOLIDATING ASSIGNMENTS"), A FOLDER containing all your Individual Consolidating Assignment deliverables.

Note: again, please keep in mind that the Critical Review of a paper is not just its condensed digest, but a critical analysis where you discuss its pros. and cons., its relevance within the contemporary literature, its possible future developments in terms of new lines of research, literature that enriches or refutes it, suggest a complementary bibliography, etc.

SUMMARY

CASE I: I'M A MEMBER OF A TEAM…

- I'll upload to QUERCUS (in the appropriate space reserved for that purpose called "FINAL DEVELOPMENT PROJECTS"), A FOLDER containing the following: {(i) My Team's Development Project Power Point Presentation (>25 Slides), (ii) The (commented) code developed and (iii) A 1 or 2 pages Executive Summary of the work presented} (ONE UPLOAD PER TEAM)
- I'll upload to QUERCUS (in the appropriate space reserved for that purpose called "INDIVIDUAL CONSOLIDATING ASSIGNMENTS"), A FOLDER containing MY Critical Review of 2 papers from the list below, each review consisting of: {(i) A self-contained PP (>25 slides) Presentation for each paper and (ii) A monography (length at my discretion) summarizing each Presentation}.

CASE II: I'M AN INDIVIDUAL WITHOUT A TEAM WHO DECIDED TO DO A DEVELOPMENT PROJECT…

- I'll upload to QUERCUS (in the appropriate space reserved for that purpose called "FINAL DEVELOPMENT PROJECTS"), A FOLDER containing the following: {(i) My Development Project Power Point Presentation (>25 Slides), (ii) The (commented) code developed and (iii) A 1 or2 pages Executive Summary of the work presented}
- I'll upload to QUERCUS (in the appropriate space reserved for that purpose called "INDIVIDUAL CONSOLIDATING ASSIGNMENTS"), A FOLDER containing the Critical Review of 2 papers consisting of: {(i) A self-contained PP (>25 slides) Presentation for each paper and (ii) A monography (length at my discretion) summarizing each Presentation}.

CASE III: I'M AN INDIVIDUAL WITHOUT A TEAM AND I'M UNABLE TO DO A DEVELOPMENT PROJECT…

- I'll upload to QUERCUS (in the appropriate space reserved for that purpose called "INDIVIDUAL CONSOLIDATING ASSIGNMENTS"), A FOLDER containing the Critical Review of 4 papers, each review consisting of: {(i) A self-contained PP (>25 slides) Presentation for each paper and (ii) A monography (length at my discretion) summarizing each Presentation}.

(LOOK AT THE END OF THIS DOCUMENT FOR ONE MORE ALTERNATIVE FOR THE INDIVIDUAL CONSOLIDATINGASSIGNMENT: THE BOOK CRITICAL REVIEW)

PAPERS FOR CRITICAL REVIEWS:

- Dias, Alexandra, The Economic Value of Controlling for Large Losses in Portfolio Selection(January 12, 2011). Available at SSRN: https://ssrn.com/abstract=1749360 or http://dx.doi.org/10.2139/ssrn.1749360
- Homescu, Cristian, Many Risks, One (Optimal?) Portfolio (July 28, 2014). Available atSSRN: https://ssrn.com/abstract=2473776 or http://dx.doi.org/10.2139/ssrn.2473776
- Butler, Adam and Philbrick, Mike and Gordillo, Rodrigo and Varadi, David, Adaptive AssetAllocation: A Primer (May 31, 2012). Available at SSRN: https://ssrn.com/abstract=2328254 or http://dx.doi.org/10.2139/ssrn.2328254
- Bhattacharya , Utpal and Galpin, Neal, The Global Rise of the Value-Weighted Portfolio (March2007). AFA 2007 Chicago Meetings Paper. Available at SSRN: https://ssrn.com/abstract=849627 or http://dx.doi.org/10.2139/ssrn.849627
- Keller, Wouter and Butler, Adam and Kipnis, Ilya, Momentum and Markowitz: A GoldenCombination (May 16, 2015). Available at SSRN: https://ssrn.com/abstract=2606884 or http://dx.doi.org/10.2139/ssrn.2606884
- Keller, Wouter, Momentum, Markowitz, and Smart Beta: A Tactical, Analytical and Practical Look atModern Portfolio Theory (June 13, 2014). Available at SSRN: https://ssrn.com/abstract=2450017or http://dx.doi.org/10.2139/ssrn.2450017
- Arnott, Robert D. and Hsu, Jason C. and Liu, Jun and Markowitz, Harry, Can Noise Create Size andValue Effects? (October 24, 2011). AFA 2008 New Orleans Meetings Paper. Available at SSRN: https://ssrn.com/abstract=936272 or http://dx.doi.org/10.2139/ssrn.936272
- Sarwar, G and Mateus, Cesario and Todorovic, Natasa, US Sector Rotation with Five-Factor Fama-French Alphas (June 16, 2017). Available at SSRN: https://ssrn.com/abstract=2987819
- Roncalli, Thierry, Keep Up the Momentum (December 7, 2017). Available atSSRN: https://ssrn.com/abstract=3083921
- Jusselin, Paul and Lezmi, Edmond and Malongo, Hassan and Masselin, Côme and Roncalli, Thierry and Dao, Tung-Lam, Understanding the Momentum Risk Premium: An In-Depth Journey Through Trend-Following Strategies (September 24, 2017). Available at SSRN: https://ssrn.com/abstract=3042173 or http://dx.doi.org/10.2139/ssrn.3042173
- Hurst, Brian and Ooi, Yao Hua and Pedersen, Lasse Heje, A Century of Evidence on Trend-Following Investing (June 27, 2017). Available at SSRN: https://ssrn.com/abstract=2993026 or http://dx.doi.org/10.2139/ssrn.2993026
- Ross, Adrienne and Moskowitz, Tobias J. and Israel, Ronen and Serban, Laura, Implementing Momentum: What Have We Learned? (December 1, 2017). Available at SSRN: https://ssrn.com/abstract=3081165 or http://dx.doi.org/10.2139/ssrn.3081165
- Mohanram, Partha S., Separating Winners from Losers Among Low Book-to-Market Stocks Using Financial Statement Analysis (April 2004). Available at SSRN: https://ssrn.com/abstract=403180 or http://dx.doi.org/10.2139/ssrn.403180
- Beneish, Messod Daniel and Nichols, Craig and Lee, Charles M.C., To Catch a Thief: Can ForensicAccounting Help Predict Stock Returns? (July 27, 2011). Available at SSRN: https://ssrn.com/abstract=1903593 or http://dx.doi.org/10.2139/ssrn.1903593

- Till, Hilary and Eagleeye, Joseph, Commodity Futures Trading Strategies: Trend-Following and Calendar Spreads (January 17, 2017). Available at SSRN: https://ssrn.com/abstract=2942340 or http://dx.doi.org/10.2139/ssrn.2942340
- Ahmerkamp, Jan Danilo and Grant, James, The Returns to Carry and Momentum Strategies (April 7,2013). Available at SSRN: https://ssrn.com/abstract=2227387 or http://dx.doi.org/10.2139/ssrn.2227387
- Baltas, Nick, Optimising Cross-Asset Carry (May 15, 2017). "Factor Investing", Elsevier & ISTE Press, 2017 (Forthcoming). Available at SSRN: https://ssrn.com/abstract=2968677
- Li, Xin and Shawky, Hany A., The Market Timing Skills of Long/Short Equity Hedge Fund Managers (July 23, 2013). Available at SSRN: https://ssrn.com/abstract=2221479 or http://dx.doi.org/10.2139/ssrn.2221479
- Lu, Chensheng and Hwang, Soosung, Cross-Sectional Stock Returns in the UK Market: The Role of Liquidity Risk. Available at SSRN: https://ssrn.com/abstract=969809
- Li, Junye, Option-Implied Volatility Factors and the Cross-Section of Market Risk Premia (May 12,2011). Available at SSRN: https://ssrn.com/abstract=1539357 or http://dx.doi.org/10.2139/ssrn.1539357
- Keller, Wouter and Keuning, Jan Willem, Breadth Momentum and Vigilant Asset Allocation (VAA): Winning More by Losing Less (July 14, 2017). Available at SSRN: https://ssrn.com/abstract=3002624
- Fang, Jiali and Qin, Yafeng and Jacobsen, Ben, Technical Market Indicators: An Overview (June 12,2014). Available at SSRN: https://ssrn.com/abstract=2449344 or http://dx.doi.org/10.2139/ssrn.2449344
- Do, Binh Huu and Faff, Robert W., Cointegration and Relative Value Arbitrage (August 19, 2016).Available at SSRN: https://ssrn.com/abstract=2785045
- Jacobs, Heiko and Weber, Martin, On the Determinants of Pairs Trading Profitability (December 22,2014). Journal of Financial Markets, Forthcoming. Available at SSRN: https://ssrn.com/abstract=2545440
- Rad, Hossein and Low, Rand Kwong Yew and Faff, Robert W., The Profitability of Pairs Trading Strategies: Distance, Cointegration, and Copula Methods (June 3, 2015). Rad, Hossein, Low, Rand Kwong Yew and Faff, Robert W., The Profitability of Pairs Trading Strategies: Distance, Cointegration, and Copula Methods, Quantitative Finance, DOI: org/10.1080/14697688.2016.1164337. Available at SSRN: https://ssrn.com/abstract=2614233 or http://dx.doi.org/10.2139/ssrn.2614233
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- Goodman, Theodore H. and Neamtiu, Monica and Zhang, Frank, Fundamental Analysis and Option Returns (January 3, 2013). Available at SSRN: https://ssrn.com/abstract=1974753 or http://dx.doi.org/10.2139/ssrn.1974753
- Tosi, Adriano and Ziegler, Alexandre, The Timing of Option Returns (March 16, 2017). Available at SSRN: https://ssrn.com/abstract=2909163 or http://dx.doi.org/10.2139/ssrn.2909163
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Momentum (March23, 2015). Journal of Investment Management, Forthcoming. Available at
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SSRN: https://ssrn.com/abstract=1363476 or http://dx.doi.org/10.2139/ssrn.1363476
- Cakici, Nusret and Tan, Sinan, Size, Value, and Momentum in Developed Country Equity Returns:Macroeconomic and Liquidity Exposures (November 19, 2013). Available at
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- De Groot, Wilma and Pang, Juan and Swinkels, Laurens, The Cross-Section of Stock Returns in Frontier Emerging Markets (August 1, 2012). Journal of Empirical Finance, Vol. 19, Issue 5, pp. 796-818.. Available at
SSRN: https://ssrn.com/abstract=1600023 or http://dx.doi.org/10.2139/ssrn.1600023
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- A Trading Approach to Testing for Predictability. Stanislav Anatolyev. New Economic School, Moscow, 117418, Russia (sanatoly@nes.ru). Alexander Gerko. Deutsche Bank, Moscow, 127051,Russia
- Volatility Prediction and Risk Management: An SVR-GARCH Approach; Gaygısız, Esma The journal of financial data science, 2020, Vol.2 (4), p.85-104; Pageant Media US. Author: Karasan, Abdullah; DOI: 10.3905/jfds.2020.1.046; Available at UofT:
https://librarysearch.library.utoronto.ca/discovery/fulldisplay?docid=cdi_crossref_primary_10_3905_jfds_2020_1_046&context=PC&vid=01UTORONTO_INST:UTORONTO&lang=en&search_scope=UTL_AND_CI&adaptor=Primo%20Central&tab=Everything&query=any,contains,Abdullah%20Karasan%20Volatility&offset=0
- High-Frequency Trading with Machine Learning Algorithms and Limit Order Book Data, Data science in finance and economics, 2022, Author: Manveer Kaur Mangat; DOI: 10.3934/dsfe.2022022; Available at UofT:
https://librarysearch.library.utoronto.ca/discovery/fulldisplay?docid=cdi_unpaywall_primary_10_3934_dsfe_2022022&context=PC&vid=01UTORONTO_INST:UTORONTO&lang=en&search_scope=UTL_AND_CI&adaptor=Primo%20Central&tab=Everything&query=any,contains,Data%20science%20finance&pfilter=rtype,exact,articles&offset=0
- Textual analysis and machine leaning: Crack unstructured data in finance and accounting; The Journal of finance and data science; Author: Guo, Li; DOI: 10.1016/j.jfds.2017.02.001; availableat UofT:
https://librarysearch.library.utoronto.ca/discovery/fulldisplay?docid=cdi_unpaywall_primary_10_1016_j_jfds_2017_02_001&context=PC&vid=01UTORONTO_INST:UTORONTO&lang=en&search_scope=UTL_AND_CI&adaptor=Primo%20Central&tab=Everything&query=any,contains,Data%20science%20finance&pfilter=rtype,exact,articles&offset=0
- Financial news predicts stock market volatility better than close price; The Journal of finance and data science; Author: Adam Atkins; DOI: 10.1016/j.jfds.2018.02.002
https://librarysearch.library.utoronto.ca/discovery/fulldisplay?docid=cdi_unpay

wall_primary_10_1016_j_jfds_2018_02_002&context=PC&vid=01UTORONTO_INST:UTORONTO&lang=en&search_scope=UTL_AND_CI&adaptor=Primo%20Central&tab=Everything&query=any,contains,Data%20science%20finance%20markets&pfilter=rtype,exact,articles&offset=0

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DEVELOPMENT PROJECTS:

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PROJECT 1:

By imitating the way it is done in the Basic Regression Channel Program, incorporate the White's Reality Check Test to the output of the Pairs Trading Program and show the advantages of this addition.

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PROJECT 2 (IMPORTANT):

Implement a risk management system by laddering the asset allocation along the time dimension and show its impact on the equity curve's volatility. It involves the of coding the "laddering" risk management system used to "atomize" risk for the ETFs momentum trading system discussed in class: Ex.: for Holding Periods of 2 Weeks, 4 Weeks, etc., respectively:

Suppose that the program that optimizes the holding period for the momentum trading system tells us that the optimal holding period for our list of ETFs is 15 days (i.e., the time that these ETFs need to develop their full momentum), but let suppose that our risk management team tells us that holding a position for 15 days exposes us to too much risk of crashing of the asset we're invested in.

How do we solve such a paradox?

If you allocate your investment capital in two portions of 50% each as explained below, every week 50% of the portfolio will get re-balanced without losing momentum! This means, among other things, that: (i) Even though we'll be rebalancing the portfolio every week, every asset will still have 2 weeks to develop its momentum (ii) Week after week the portfolio will be fully diversified (iii) If there is a crash in one of the assets (A, B, C,…, G), you'll be risking only a portion of your invested capital because you will never be fully invested in any of the assets:

On week 1 we invest the first 50% in ETF "A" and hold A for 2 weeks	On week 2 we invest the second 50% in ETF "B" and hold B for 2 weeks	On week 3 we re-invest the 50% that was in A (available again because 2 weeks have already passed) in ETF "C" and hold C for 2 weeks	On week 4 we re-invest the 50% that was in B (available again because 2 weeks have already passed) in ETF "D" and hold D for 2 weeks	On week 5 we re-invest the 50% that was in C (which is available again) in ETF "E" and hold E for 2 weeks	On week 6 we re-invest the 50% that was in D (which is available again) in ETF "F" and hold F for 2 weeks	On week 7 we re-invest the 50% that was in E (which is available again) in ETF "G" and hold G for 2 weeks
Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7
A	A	C	C	E	E	G
	B	B	D	D	F	F

We proceed in a similar way for the case in which the optimal momentum-recommended holding period is 1 month, which being even longer, is even riskier. The risk of holding an investment for 1 month can be "Laddered" in a similar fashion using 25% portions of the capital, which allows us to, again, rebalance the portfolio every week and effectively reduce our risk without sacrificing momentum.

If Capital is Allocated in Four Portions (25% Each) => Every Week 25% of the Portfolio gets Re-Balanced

A	B	C	D	E	F	G	H	I	J
A	A	A	A	E	E	E	E	I	I
	B	B	B	B	F	F	F	F	J
		C	C	C	C	G	G	G	G
			D	D	D	D	H	H	H

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PROJECT 3 (UPGRADE OF PROJECT 2):

Adapt the Rotational Momentum Program to trade automatically (and independently) a bond ETF portfolio, a stock ETF portfolio (and perhaps 2 or 3 more portfolios), automatically integrating all the obtained equity curves as a weighted average.

Ex.: to trade a bond ETF portfolio like ["BIL","TIP","IEI","IEF","TLH","TLT","SHY"] (different maturities Govt. Bonds) and a stock ETF portfolio like ["XLB","XLE","XLF", "XLI", "XLK", "XLP", "XLU", "XLV", "XLY", "TLT"] (10 sectors of the SPY), and then integrate the two equity curves into a single one.

PROJECT 3 **MUST** INCLUDE THE RISK-MANAGEMENT SYSTEM IN 2.

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PROJECT 4:

Use Hierarchical (or any other appropriate kind of) Clustering, possibly with Returns & Volatility as Metrics, to find [ETF families of "Congruent" ETFs] that perform as well under the Rotational Momentum Program], as [the 7 Govt. Bonds ETFs] or [the 21 Industrial Sectors ETFs] that we proposed in the class on Rotational Momentum.

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PROJECT 5: incorporating the V-RATIO to the MOMENTUM trading methodology: use it to determine the STRENGTH of the trend and therefore "Refine" (improve) the entry-rule by adding this condition to it. Consider substituting the score of the ETFs by the V-RATIO or reinforcing the "Score" by adding the V-RATIO to the score.

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PROJECT 6: Design a back-testing algorithm for the credit spread arbitrage detection strategy discussed in the options and derivatives part of the course. Code and back-test it.

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PROJECT 7: Use Johnson's Distribution to add to the Single Asset Trading Regression Channel to determine the "fences" with greater precision (This distribution mimics a stock much better than the Normal Distribution)

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PROJECT 8: Use GARCH as a Volatility Predictor in order to better determine the "fences" of the Single Asset Trading Regression Channel Trading Program

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PROJECT 9: Add White's Reality Check to the Loop Pairs Trading Program to be able to distinguish (qualify) and select the "good" Equity Curves.

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PROJECT 10: To design a "STABLE COIN" based on the trading of the pair BTC/US$ v.s. BTC/EUR with

the Pairs Trading Program.

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PROJECT X: any project proposed by a team that is discussed and accepted as viable by the instructors

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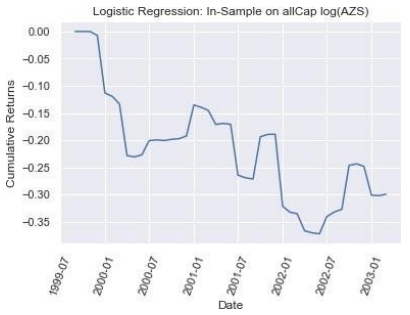
POSSIBLE PROJECTS RELATED TO

THE"ALTMAN-BASED" CRASH

INDICATOR

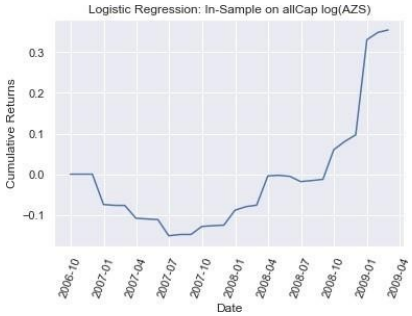
2001 TECH BUBBLE CRASH



UPPER CHART: ALTMAN-BASED C. I.

LOWER CHART: S&P 500

2007-2008 MARKET CRASH



UPPER CHART: ALTMAN-BASED C. I.

LOWER CHART: S&P 500

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POSSIBLE "VALUE" DEVELOPMENT PROJECTS:

1) TO REFINE THE CRASH INDICATOR (WHICH IS: LONG IF ALTMAN > 3, SHORT IF ALTMAN < 3) BY ADDING A COMPLEMENTARY INDICATOR THAT TELLS WHAT PERCENTAGE (%) OF THE COMPANIES ARE AT RISK (= HOW MANY HAVE ALTMAN < 3); PERHAPS THIS WOULD MAKE THE

INDICATOR “MORE SENSITIVE” AND ABLE TO ‘QUANTIFY” AT WHAT PERCENTAGE OF “BROKEN” COMPANIES THECRASH INDEX STARTS CLIMBING AND THE MARKET STARTS FALTERING

2) ALLOW FOR THE TRESHOLD (= 3) TO BE CHANGED TO 1.87 WHERE THE COMPANIES ARE NOT JUST AT RISK, BUT JUST A COUPLE OF MONTHS AWAY FROM BANKRUPTCY.
3) TO CREATE AN EXACTLY “ANALOGOUS” PIOTROSKI CRASH INDICATOR USING THE PIOTROSKI INDEX WITH 4 AS THRESHOLD IN PLACE OF THE ALTMAN INDEX WITH 3 AS THRESHOLD: PIOTROSKI < 4 USUALLY INDICATES DECREASE IN PRICE BUT NOT BANKRUPTCY; SO, POSSIBLY PIOTROSKI WILL DETECT THE BEGINNING OF A DIP AND ALTMAN WILL CONFIRM (OR REFUTE) ITS GRAVITY AS A CRASH.
4) TO EXPLORE THEPOSSIBLE SYNERGIES BETWEEN 2) AND 3) USING A.I.
5) CALCULATE THE “PIOTROSKI SCORE” AND THE “ALTMAN SCORE” OF AN ETF (BY USING THE PROGRAM AND THE DATA ACCESS WE PROVIDED), USING THE 10 MAIN STOCKS THAT “CARRY” THE MAIN WEIGHT OF EACH ETF, FOR ALL THE 25 STOCKS REPRESENTING THE 25 MAIN INDUSTRIAL SECTORS OF THE MARKET.

A FEW ADDITIONAL POSSIBLE DEVELOPMENT PROJECTS (INTEGRATING VALUE, MOMENTUM AND RISK MANAGEMENT IN A UNIFIED STRATEGY)

6) ONCE 5) IS DONE, WE CAN BUILD A PORTFOLIO CONSISTING OF THESE 25 [PIOTROSKI-ALTMAN QUALIFIED ETFS] (INSTEAD OF 25 STOCKS), AND TRADE EACH OF THESE ETFS (THE COMPONENTS OF THE PORTFOLIO) USING THE REGRESSION CHANNEL, AS YOU WOULD DO WITH 25 [PIOTROSKI-ALTMAN QUALIFIED STOCKS].
7) NOTICE THAT ONE OF THE ADVANTAGES OF 6) IS THAT WHEN TRADING THESE 25 ETFS, YOU CAN HAVE THE INPUT OF THE ETF ROTATIONAL MOMENTUM PROGRAM THAT ALSO TRADES THESE 25 ETFS; OR MAY BE BETTER TO SAY THAT WHILE TRADING WITH THE ROTATIONAL MOMENTUM PROGRAM, EVERY TIME AN ETF IS CHOSEN BY THIS PROGRAM, BEFORE INVESTING IN THE ETF YOU CAN VERYFY ITS [PIOTROSKI-ALTMAN QUALIFICATION] AS WELL AS ITS OVERBOUGHT-OVERSOLD CONDITION REGARDING THE REGRESSION CHANNEL.
8) REMEMBER THAT WE USE 2) AND 3) TO DETERMINE WHEN THE MARKET (SPX) IS GOING DOWN IN ORDER TO BUY [AN “AT THE MONEY” 45 TO 65 DAYS DEEP PUT OPTION ON THE SPX] THAT WILL COVER POSSIBLE LOSSES IN THE PORTFOLIO IF THERE IS A CRASH IN THE MARKET.
9) ALSO REMEMBER THAT AN INDEPENDENT RISK MANAGEMENT SYSTEM BASED IN THE “LADDERING” OF [THE 25 ETFS TRADED WITH THE MOMENTUM PROGRAM] SHOULD BE IN PLACEAT THE SAME TIME.

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ONE MORE ALTERNATIVE FOR THE INDIVIDUAL CONSOLIDATING ASSIGNMENT:

FOR THE INDIVIDUAL CONSOLIDATING ASSIGNMENT, instead of 2 (two) paper reviews YOU CAN DO 1 (ONE) BOOK REVIEW based on ONE of the 5 books listed below. All of them are great books that will enrich your personal finance library, so we definitely encourage you to buy them. Nevertheless, we've been told (haven't verified it) that there is a place called z-library or something of the sort where you can go and look at any book to check if its content coincides with your interests and expectations before buying it.

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BOOKS FOR CRITICAL REVIEWS:

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- "Dual Momentum Investing: An Innovative Strategy for Higher Returns with LowerRisk". Editor: McGraw-Hill; Author: Gary Antonacci
- "The Little Book That Still Beats the Market". Editor: Wiley; Author: Joel Greenblatt
- "The Ivy Portfolio: How to Invest Like the Top Endowments and Avoid BearMarkets". Editor: Wiley; Authors: Mebane T. Faber, Eric W. Richardson
- "DIY Financial Advisor: A Simple Solution to Build and Protect Your Wealth". Editor: Wiley; Authors: Wesley R. Gray, Ph.D.; Jack R. Vogel, Ph.D.; David P. Foulke, Ph.D..
- "Evidence-Based Technical Analysis". Editor: Wiley; Author: David Aronson.
- Permutation and Randomization Tests for Trading System Development: Algorithms in C++ Paperback – Feb. 12 2020; Author: Timothy Masters.
- Statistically Sound Indicators For Financial Market Prediction: Algorithms in C++ Paperback – Oct. 22 2019: Author: Timothy Masters
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